

Operational Semantics

As explained in the untyped λ -calculus lecture, the Church encoding of Booleans is as follows. The Boolean **true** is defined as $\lambda x.\lambda y.x$, and the Boolean **false** is defined as $\lambda x.\lambda y.y$. The negation function, written $\neg$ here, is defined as follows: $\lambda a.a \text{ false true}$.

- Show that $\neg \text{true}$ reduces to **false** using the call-by-value small-step operational semantics (using evaluation contexts).
- Show that $\neg \text{true}$ reduces to **false** using the call-by-value big-step operational semantics.
- Define an **xor** function that takes two arguments and returns true iff the two arguments are different.
- Show that **xor true false** reduces to **true** using the call-by-value small-step operational semantics (using evaluation contexts).
- Show that **xor true true** reduces to **false** using the call-by-value small-step operational semantics (using evaluation contexts).
- Show that **xor true false** reduces to **true** using the call-by-value big-step operational semantics.
- Show that **xor true true** reduces to **false** using the call-by-value big-step operational semantics.

Solution

- $\neg \text{true}$ reduces to false using the call-by-value small-step operational semantics as follows:

$$\begin{array}{c}
 \text{---} \\
 \frac{}{\neg \text{true} \rightarrow_v \text{true false true}} \beta \\
 \text{---} \\
 \frac{\frac{}{\text{true false} \rightarrow_v \lambda y. \text{false}} \beta}{\text{true false true} \rightarrow_v (\lambda y. \text{false}) \text{true}} \text{APP} \bullet \text{true} \\
 \text{---} \\
 \frac{}{(\lambda y. \text{false}) \text{true} \rightarrow_v \text{false}} \beta
 \end{array}$$

- $\neg \text{true}$ reduces to false using the call-by-value big-step operational semantics as follows:

$$\frac{\frac{}{\neg \Rightarrow_v \neg} \text{VAL} \quad \frac{}{\text{true} \Rightarrow_v \text{true}} \text{VAL} \quad \Pi \quad \frac{\frac{}{\text{true} \Rightarrow_v \text{true}} \text{VAL} \quad \frac{}{\text{false} \Rightarrow_v \text{false}} \text{VAL}}{\text{true false true} \Rightarrow_v \text{false}} \text{APP} \quad \frac{}{\neg \text{true} \Rightarrow_v \text{false}} \text{APP}$$

where Π is

$$\frac{\frac{}{\text{true} \Rightarrow_v \text{true}} \text{VAL} \quad \frac{}{\text{false} \Rightarrow_v \text{false}} \text{VAL} \quad \frac{}{\lambda y. \text{false} \Rightarrow_v \lambda y. \text{false}} \text{VAL}}{\text{true false} \Rightarrow_v \lambda y. \text{false}} \text{APP}$$

- The xor function can be defined as follows:

$$\lambda a. \lambda b. a (\neg b) b$$

- xor true false reduces to true using the call-by-value small-step operational semantics as follows:

$$\begin{array}{c}
 \text{---} \\
 \frac{\frac{}{\text{xor true} \rightarrow_v \lambda b. \text{true} (\neg b) b} \beta}{\text{xor true false} \rightarrow_v (\lambda b. \text{true} (\neg b) b) \text{false}} \text{APP} \bullet \text{false} \\
 \text{---} \\
 \frac{}{(\lambda b. \text{true} (\neg b) b) \text{false} \rightarrow_v \text{true} (\neg \text{false}) \text{false}} \beta \\
 \text{---} \\
 \frac{\frac{}{\neg \text{false} \rightarrow_v \text{false false true}} \beta}{\text{true} (\neg \text{false}) \text{false} \rightarrow_v \text{true} (\text{false false true}) \text{false}} \text{APP}_{\text{true}} \bullet \text{false} \\
 \text{---} \\
 \frac{\frac{}{\text{false false} \rightarrow_v \lambda y. y} \beta}{\text{true} (\text{false false true}) \text{false} \rightarrow_v \text{true} ((\lambda y. y) \text{true}) \text{false}} \text{APP}_{\text{true}} (\bullet \text{true}) \text{false} \\
 \text{---} \\
 \frac{\frac{}{(\lambda y. y) \text{true} \rightarrow_v \text{true}} \beta}{\text{true} ((\lambda y. y) \text{true}) \text{false} \rightarrow_v \text{true true false}} \text{APP}_{\text{true}} \bullet \text{false} \\
 \text{---} \\
 \frac{\frac{}{\text{true true} \rightarrow_v \lambda y. \text{true}} \beta}{\text{true true false} \rightarrow_v (\lambda y. \text{true}) \text{false}} \text{APP} \bullet \text{false} \\
 \text{---} \\
 \frac{}{(\lambda y. \text{true}) \text{false} \rightarrow_v \text{true}} \beta
 \end{array}$$

- `xor true true` reduces to `false` using the call-by-value small-step operational semantics as follows:

$$\begin{array}{c}
 \text{---} \\
 \frac{\frac{\text{---}}{\text{xor true} \rightarrow_v \lambda b.\text{true} (\neg b) b} \beta}{\text{xor true true} \rightarrow_v (\lambda b.\text{true} (\neg b) b) \text{true}} \text{APP}_{\bullet \text{ true}} \\
 \text{---} \\
 \frac{\text{---}}{(\lambda b.\text{true} (\neg b) b) \text{true} \rightarrow_v \text{true} (\neg \text{true}) \text{true}} \beta \\
 \text{---} \\
 \frac{\frac{\text{---}}{\neg \text{true} \rightarrow_v \text{true false true}} \beta}{\text{true} (\neg \text{true}) \text{true} \rightarrow_v \text{true} (\text{true false true}) \text{true}} \text{APP}_{\text{true} \bullet \text{ true}} \\
 \text{---} \\
 \frac{\frac{\text{---}}{\text{true false} \rightarrow_v \lambda y.\text{false}} \beta}{\text{true} (\text{true false true}) \text{true} \rightarrow_v \text{true} ((\lambda y.\text{false}) \text{true}) \text{true}} \text{APP}_{\text{true} (\bullet \text{ true}) \text{ true}} \\
 \text{---} \\
 \frac{\frac{\text{---}}{(\lambda y.\text{false}) \text{true} \rightarrow_v \text{false}} \beta}{\text{true} ((\lambda y.\text{false}) \text{true}) \text{true} \rightarrow_v \text{true false true}} \text{APP}_{\text{true} \bullet \text{ true}} \\
 \text{---} \\
 \frac{\frac{\text{---}}{\text{true false} \rightarrow_v \lambda y.\text{false}} \beta}{\text{true false true} \rightarrow_v (\lambda y.\text{false}) \text{true}} \text{APP}_{\bullet \text{ true}} \\
 \text{---} \\
 \frac{\text{---}}{(\lambda y.\text{false}) \text{true} \rightarrow_v \text{false}} \beta
 \end{array}$$

- `xor true false` reduces to `true` using the call-by-value big-step operational semantics as follows:

$$\frac{\Pi_0 \quad \frac{\text{---}}{\text{false} \Rightarrow_v \text{false}} \text{VAL} \quad \frac{\Pi_1 \quad \frac{\text{---}}{\text{false} \Rightarrow_v \text{false}} \text{VAL} \quad \frac{\text{---}}{\text{true} \Rightarrow_v \text{true}} \text{VAL}}{\text{true} (\neg \text{false}) \text{false} \Rightarrow_v \text{true}} \text{APP}}{\text{xor true false} \Rightarrow_v \text{true}} \text{APP}$$

where Π_0 is

$$\frac{\frac{\text{---}}{\text{xor} \Rightarrow_v \text{xor}} \text{VAL} \quad \frac{\text{---}}{\text{true} \Rightarrow_v \text{true}} \text{VAL} \quad \frac{\text{---}}{\lambda b.\text{true} (\neg b) b \Rightarrow_v \lambda b.\text{true} (\neg b) b} \text{VAL}}{\text{xor true} \Rightarrow_v \lambda b.\text{true} (\neg b) b} \text{APP}$$

where Π_1 is

$$\frac{\frac{\text{---}}{\text{true} \Rightarrow_v \text{true}} \text{VAL} \quad \frac{\neg \Rightarrow_v \neg \text{VAL} \quad \frac{\text{---}}{\text{false} \Rightarrow_v \text{false}} \text{VAL} \quad \Pi_2}{\neg \text{false} \Rightarrow_v \text{true}} \text{APP} \quad \frac{\text{---}}{\lambda y.\text{true} \Rightarrow_v \lambda y.\text{true}} \text{VAL}}{\text{true} (\neg \text{false}) \Rightarrow_v \lambda x.\text{true}} \text{APP}$$

where Π_2 is

$$\frac{\frac{\text{---}}{\text{false} \Rightarrow_v \text{false}} \text{VAL} \quad \frac{\text{---}}{\text{false} \Rightarrow_v \text{false}} \text{VAL} \quad \frac{\text{---}}{\lambda y.y \Rightarrow_v \lambda y.y} \text{VAL}}{\text{false false} \Rightarrow_v \lambda y.y} \text{APP} \quad \frac{\text{---}}{\text{true} \Rightarrow_v \text{true}} \text{VAL} \quad \frac{\text{---}}{\text{true} \Rightarrow_v \text{true}} \text{VAL}}{\text{false false true} \Rightarrow_v \text{true}} \text{APP}$$

- `xor true true` reduces to `false` using the call-by-value big-step operational semantics as follows:

$$\frac{\Pi_0 \quad \overline{\text{true} \Rightarrow_v \text{true}} \text{ VAL} \quad \frac{\Pi_1 \quad \overline{\text{true} \Rightarrow_v \text{true}} \text{ VAL} \quad \overline{\text{false} \Rightarrow_v \text{false}} \text{ VAL}}{\overline{\text{true} (\neg \text{true}) \text{true} \Rightarrow_v \text{false}} \text{ APP}} \text{ APP} \quad \overline{\text{xor true true} \Rightarrow_v \text{false}}$$

where Π_0 is

$$\frac{\overline{\text{xor} \Rightarrow_v \text{xor}} \text{ VAL} \quad \overline{\text{true} \Rightarrow_v \text{true}} \text{ VAL} \quad \overline{\lambda b. \text{true} (\neg b) b \Rightarrow_v \lambda b. \text{true} (\neg b) b} \text{ VAL}}{\overline{\text{xor true} \Rightarrow_v \lambda b. \text{true} (\neg b) b} \text{ APP}}$$

where Π_1 is

$$\frac{\overline{\text{true} \Rightarrow_v \text{true}} \text{ VAL} \quad \overline{\neg \Rightarrow_v \neg} \text{ VAL} \quad \overline{\text{true} \Rightarrow_v \text{true}} \text{ VAL} \quad \Pi_2 \text{ APP} \quad \overline{\lambda y. \text{false} \Rightarrow_v \lambda y. \text{false}} \text{ VAL}}{\overline{\text{true} (\neg \text{true}) \Rightarrow_v \lambda x. \text{false}} \text{ APP}}$$

where Π_2 is

$$\frac{\overline{\text{true} \Rightarrow_v \text{true}} \text{ VAL} \quad \overline{\text{false} \Rightarrow_v \text{false}} \text{ VAL} \quad \overline{\lambda y. \text{false} \Rightarrow_v \lambda y. \text{false}} \text{ VAL}}{\overline{\text{true false} \Rightarrow_v \lambda y. \text{false}} \text{ APP}} \quad \overline{\text{true} \Rightarrow_v \text{true}} \text{ VAL} \quad \overline{\text{false} \Rightarrow_v \text{false}} \text{ VAL}}{\overline{\text{true false true} \Rightarrow_v \text{false}} \text{ APP}}$$